

CS 7545 F26, Computational Mini Project 1

100 point assignment

Due September 10, 2026

In this project, you'll compare several concentration inequalities for a Binomial random variable.

Let $S_n = \sum_{i=1}^n X_i$ where X_i are sampled independently according to,

$$X_i = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases} \quad (1)$$

Part 1

In this first part, you'll write Python functions (1.A, 1.B, 1.D) and prove a result (1.C). In your submission, include the code snippets and your proof.

Part 1.A (10 points)

Write a function `Hoeffding_Bound(n,p,greater_than,k)` which returns a bound on $\Pr(S_n \geq k)$ if `greater_than==True` and $\Pr(S_n \leq k)$ if `greater_than==False`. Be sure to account for the fact that Hoeffding requires $np \leq k$ to bound $\Pr(S_n \geq k)$. When $np > k$, return the trivial bound 1. Handle the lower tail analogously.

In your submission, include the Python code snippet for the written function.

Part 1.B (10 points)

Hoeffding's Inequality is quite loose for small p . We can use the fact that the variance of X_i decays for small p .

Write a function `Bernstein_Bound(n,p,greater_than,k)`. Be sure to handle $np > k$ for the upper tail bound, as in Part 1.A.

In your submission, include the Python code snippet for the written function.

Part 1.C (20 points)

Here, we'll derive and test a Chernoff bound without any algebraic approximations at the end of the proof.

The beginning of the Chernoff bound (with $t > 0$) is:

$$\Pr(S_n \geq k) = \Pr\left(\exp\left(\sum_{i=1}^n tX_i\right) \geq \exp(tk)\right) \quad (2)$$

$$\leq \frac{\mathbb{E}\left[\exp\left(\sum_{i=1}^n tX_i\right)\right]}{\exp(tk)} \quad (3)$$

$$= \frac{\mathbb{E}\left[\prod_{i=1}^n \exp(tX_i)\right]}{\exp(tk)} \quad (4)$$

$$= \frac{\prod_{i=1}^n \mathbb{E}[\exp(tX_i)]}{\exp(tk)} \quad (5)$$

$$= \frac{\prod_{i=1}^n (p \exp(t) + (1-p))}{\exp(tk)} \quad (6)$$

$$= \frac{(1 + p(\exp(t) - 1))^n}{\exp(tk)} \quad (7)$$

Continue the proof (without approximations), including optimizing over the value of t , to show that for $k \geq np$,

$$\Pr(S_n \geq k) \leq \exp(-nD_{\text{KL}}((k/n)||p)) \quad (8)$$

Where $D_{\text{KL}}(a||b) = a \ln(a/b) + (1-a) \ln((1-a)/(1-b))$ is the KL-divergence between a Bernoulli with probability a and a Bernoulli with probability b .

In your submission, include your derivation.

Part 1.D (10 points)

Write a function `Chernoff_Bound(n,p,greater_than,k)`. Be sure to account for the edge cases of $p = 0$ and $p = 1$. Be sure to handle $np > k$ for the upper tail bound, as in Part 1.A.

In your submission, include the Python code snippet for the written function.

Part 2

Here, we'll run several experiments with the functions written above. It is recommended to use the provided starter code. In your submission, include the print out and answer the questions.

Part 2.A (5 points)

With the Binomial CDF (already in starter code) and your three bounds above, compute $\Pr(S_n \geq k)$ for $n = 1000$, $p = 0.05$, and $k = 75$.

Discuss the ordering of the probabilities (smaller is tighter) and the gaps between them.

In your submission, include the program output as well as your discussion.

Part 2.B (10 points)

We can “invert” the bounds by searching for a value of p that makes the tails less than $\delta/2$. For the Binomial CDF, this is known as the Clopper-Pearson confidence interval.

Find the largest p_1 and smallest p_2 (with $\delta = 0.05$) such that

- $\Pr(S_n \geq k) \leq \delta/2$ for $p = p_1$
- $\Pr(S_n \leq k) \leq \delta/2$ for $p = p_2$

For the following values of n and k ,

- $n = 1000, k = 100$
- $n = 1000, k = 500$
- $n = 1000, k = 10$
- $n = 1000, k = 0$
- $n = 100, k = 10$
- $n = 10000, k = 1000$

Discuss the relative sizes of the intervals (p_1, p_2) for the different methods and for different values of n and k . Also, answer:

- How do the sizes of the intervals change with n and k ?
- Do you observe that Hoeffding’s Inequality is looser for small values of k/n ?
- How do the differences in probabilities (for different bounds) from Part 2.A compare to the differences in the confidence intervals?

In your submission, include the program output as well as your discussion.

Part 2.C (15 points)

Next, we'll test how often the true value of p falls in the confidence interval. For $n = 1000$ and $p = 0.1$, sample S_n , then compute a confidence interval using $\delta = 0.05$. For a million samples, how often does p lie outside the confidence interval?

- Compute the miss rate (i.e. the confidence interval misses p) for each of the 4 bounds.
- Choose different values for n , p , and δ , and compute the miss rate for each of the 4 bounds.
- Discuss how the miss rate changes for different bounds.

In your submission, include the program output as well as your discussion.

Part 2.D (20 points)

Finally, plot the confidence intervals while varying k and varying n .

First, fix $n = 1000$. Plot $0 \leq k \leq n$ on the x-axis and the confidence intervals on the y-axis.

Next, fix $n/k = 10$ and vary $1 \leq k \leq 1000$. Plot n on the x-axis and the confidence intervals on the y-axis. You will need to edit the starter code for this plot.

For each plot, discuss how the four methods compare for different values of k and n . Is the ordering of methods always the same? Is the gap larger for some values of n or k ?

In your submission, include the two plots as well as your discussion.